Differential Expression Analysis of the transcriptome of human Trabecular meshwork cells stimulated with Transforming Growth Factor β-1 (TGF-β1) involved in Pseudoexfoliation Syndrome (XFS) and Glaucoma (XFG)

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ABSTRACT

Pseudoexfoliation Syndrome (XFS) is a chronic disorder characterised by the accumulation of extracellular material in various body tissues, primarily ocular tissues, increasing intraocular pressure. Affecting an estimated 60 million people worldwide, XFS can eventually evolve to Pseudoexfoliation Glaucoma (XFG), a more severe condition. The molecular mechanisms underlying XFS/XFG are not clearly understood; yet increasing evidence suggests that the Transforming Growth Factor Beta 1 (TGF-β1) signalling pathway plays a significant role in the disease's development.

This article aims to understand which genes are related to XFG and to identify which biological processes are potentially activated or inhibited in this pathology. To achieve the aim, computational tools such as National Centre for Biotechnology Information (NCBI) and Protein Data Bank (PDB) allowed the global characterisation of TGF-β1. Using RNA sequencing, the human trabecular meshwork cells' transcriptome was analysed after stimulation with TGF-β1 using RStudio® and Galaxy European Server. This approach enables the identification of differentially expressed genes and biological functions related to XFS/XFG.

Thanks to the differential expression (DE) and functional enrichment analysis, ITGB6 seems to be related to XFG pathology. A deeper understanding of ITGB6's functions and mechanisms may enable its use as a biomarker or therapeutic target for managing the pathological ECM changes associated in XFG.

Key words: ***Pseudoexfoliation Syndrome (XFS), Pseudoexfoliation Glaucoma (XFG), Transforming Growth Factor Beta 1 (TGF-β1), Transforming Growth Factor Receptor beta (TGFR-β1), Integrin Subunit Beta 6 (ITGB6)& Differential analysis (DE).***

1. INTRODUCTION

Pseudoexfoliation syndrome (XFS) is an age-related and chronic disorder characterised by the accumulation of extracellular material in various body tissues. This condition primarily affects ocular tissues, especially in the eye's anterior segment, increasing intraocular pressure (Tuteja et al., 2023). XFS is estimated to affect 60 million people worldwide (Aboobakar et al., 2017), of whom half will eventually develop Pseudoexfoliation Glaucoma (XFG), which is one of the major causes of blindness globally (Tuteja et al., 2023).

XFG is an aggressive form of secondary open-angle glaucoma, characterised by the accumulation of exfoliation material (XFM). The XFM contains elastin, laminin, fibronectin, cross-linking enzyme LOXL1 and latent Transforming Growth Factor Beta (TGF-β) binding proteins, among others. It has been observed that activation of the TGF-β pathway by Transforming Growth Factor Beta 1 (TGF-β1) ligand has been implicated in the pathogenesis of XFG (TGFBR1 Gene: MedlinePlus Genetics, n.d.). Moreover, the TGF-β1 is also known to be up-regulated in this pathology.

The activation of the TGF-β pathway not only affects XFG but is also crucial for regulating various cellular functions, such as cell division, proliferation, differentiation, movement, death and inflammatory response, since TGF-β1 belongs to the TGF-β superfamily of cytokines (MedlinePlus Genetics, n.d.) (Park et altreated., 2021).

The transcriptome data used in this study was obtained from the samples of the "Genome-Wide RNA Sequencing of Human Trabecular Meshwork Cells Treated with TGF-β1: Relevance to Pseudoexfoliation Glaucoma" study by Roodnat AW et.al. The eight samples consist of normal human trabecular meshwork (HTM) eye cells. Four of them, the Treatment samples were stimulated with recombinant human TGF-β1 and the other four, the Controls, were not stimulated. The complete transcriptome of these cells was extracted and sequenced.

This article aims to understand which genes, from the sample's transcriptome, are related to the XFG and to identify which biological processes are potentially activated in this pathology. First, a global characterisation of both the TGF-β1 and its receptor (TGFR-β1) gene and protein sequences were done. Their location, size, and function, among others, were determined. Moreover, a search for orthologs and a phylogenetic tree was conducted using the database Ensembl and the online tool Phylo.io (included inside Ensembl). The complete information, with the detailed steps about this initial gene sequence and protein characterisation, can be found in **Annex A**.

Then, all the cells' transcriptome data was analysed to investigate their impact on XFG. The analysis was conducted using RStudio®, combined with online tools such as the Galaxy European server, NCBI and g:Profiler, among others. A differential expression (DE) analysis was conducted to detect the differentially expressed genes within the Treatment group. Next, several GO enrichments were performed to determine which biological processes are related to XFG.

By understanding which genes and biological processes are involved in XFG, not only can the aetiology of the disease be identified, but also, more effective treatments and potential preventive medicine could be developed. This is particularly significant, as XFG remains one of the leading causes of blindness.

2. MATERIALS AND METHODS

2.1. Gene and protein characterisation

To gather detailed information about the specific human ligand and its receptor, different databases were used, such as NCBI, BLAST, Uniprot and Ensembl. The information obtained about these proteins can be found in **Annex A**.

2.2. RNA-seq analysis with Galaxy

RNA sequencing (RNA-seq) data was analysed using the Galaxy European server. This platform allowed access to the Sequence Read Archive (SRA) and enabled the processing of high-throughput sequencing data. Raw data in FASTQ format was assessed for quality using MultiQC, which allowed an overview of sample quality metrics.

After the initial quality assessment, the sequencing reads were trimmed on the Galaxy European server. This step involved selecting a collection of FASTQ files, removing adapter sequences, setting quality score thresholds to a minimum length of twenty nucleotides. All sequences below the threshold were discarded. Post-trimming, FastQC and MultiQC were rerun to evaluate improvements in read quality.

Following trimming and quality reassessment using HISAT2, reads were mapped to the selected reference genome, which was Genome Reference Consortium Human Reference 38 (GRCh38), versioned in the human genome from December 2013. Finally, gene-level quantification was performed using featureCounts, which summarises the number of reads assigned to genomic features or meta-features, enabling downstream expression analysis.

To perform DE analysis, the DESeq2 tool was used within the Galaxy European server, which enabled the selection of proper output formats and the generation of plots for comparing expression levels across different conditions. Visualisations were generated to assess expression trends and sample clustering. Corresponding results can be found in **Annex B**. The resulting data tables were exported for further downstream analysis in RStudio®.

2.3. RNA-seq analysis with R

2.3.1. Pre-processing and global understanding of the data

RNA-seq data was analysed using RStudio® (R version 4.4.1) to profile the relative concentrations of RNA molecules across biological samples. Gene expression changes associated with specific experimental conditions (e.g., Treatment vs. Control sample) were analysed. The graphs were coloured using the library wesanderson (v.0.3.7) to enhance visual harmony.

The input data included a raw count matrix, where rows represented genes, and columns denoted samples, along with a custom metadata (found in the Data availability section). Each entry in the matrix indicated the number of sequencing reads mapped to a specific gene, reflecting its expression level. Regarding the metadata, variables Sex and Age were not considered in the analysis, because when trying to create a DEseq2 object, the code could not be run as it detected certain associations within the data. Age was useful in confirming that the data was paired, meaning that from each patient a Treatment and a Control sample were obtained. However, as all the patients were within the same age range, Age was not considered a significant variable for analysis.

In order to have a global visualisation of the expression of the data and the clustering patterns, a heatmap was conducted using the ggplot2 (v.3.5.1) and the pheatmap (v.1.0.12) packages. Considering that it consisted of paired samples, pre-data treatment and normalisation were performed to account for differences in sequencing depth and sample-specific biases (e.g., variations in RNA extraction or library preparation). The normalisation was conducted using the Counts Per Million reads (CPM) function with the edgeR package (v.4.4.2). Heatmaps display expression levels using a colour gradient, enabling the identification of global expression patterns and detection of outliers. Hierarchical clustering was incorporated via dendrograms to group genes with similar expression profiles, allowing the detection of clear differences related to the variables of interest.

For the DE analysis (explained in the 2.3.2 section), a DESeq2 object was created by combining the raw count data with the sample metadata, including experimental Condition and

Sample_IDs as covariates to account for the paired structure data. This was conducted using the DESeq2 package (v.1.46.0) in RStudio®.

Moreover, to enhance the understanding of the data, a Principal Component Analysis (PCA) was performed using the ggplot2 (v.3.5.1) package, which allowed assessing the overall structure of the dataset and examining how samples are clustered based on biological variables. To conduct the PCA, the data needed to be normalised, which was done via the rlog function, which transforms the count data to the log2 scale.

2.3.2. Differential expression analysis with DESeq2

Differential expression (DE) analysis is used to identify genes whose expression levels significantly differ between distinct experimental conditions. This process helps reveal biological changes associated with specific treatments or disease states. RNA-seq data was analysed in RStudio® using standard bioinformatics packages such as DESeq2 (v.1.46.0), which models gene counts using a negative binomial distribution. This approach enables a robust estimation of dispersion and fold changes, allowing for the reliable detection of genes differentially expressed between experimental groups.

RNA-seq count data would have to be modelled using a Poisson distribution. However, since the RNA-seq data often exhibits overdispersion, where the variance exceeds the mean, the Poisson model is inadequate. To address this, a negative binomial distribution was employed, as it better accounts for biological variability and overdispersion in count data.

To observe the results from the DE analysis, a volcano plot was generated using the library EnhancedVolcano (v.1.24.0). With the results from this analysis, a data frame was created containing only the products with a p-value < 0.05, indicating a statistical significance. The obtained data is normalised once again via the CPM function to generate a heatmap. Furthermore, a list of the top three genes differentially expressed and up-regulated at the Treatment condition was performed. Later on, these genes were submitted to g:Profiler, which is a gene ID conversion tool, for annotation. The Entrez Gene IDs of these genes were introduced at the g:Convert tool (a function within g:Profiler) to obtain the name and a brief description of the genes. The organism was set to *Homo sapiens* to guarantee correct annotation. The results were then used for further functional analysis.

2.3.3. GO enrichment

Following DE analysis, significantly differentially expressed genes were subjected to functional enrichment analysis, in order to identify statistically significant over-representation of pathways within the list input of genes. All the steps explained below were performed using the topGO RStudio® package (v.2.58.0).

Before creating a topGO object, the NA (lowly expressed genes) values were erased. The topGO object allows for observing whether the number of genes associated with a pathway in the set is significantly larger than what would be expected by chance. Since multiple testing is conducted, to ensure statistical reliability, the False Discovery Rate (FDR) was used to minimise the error.

In order to do a more exhaustive analysis, GO enrichment was again performed for both up- and down-regulated genes, adding the log2FoldChange variable when filtering. Different thresholds were applied: for the up-regulated ones >+2 and >+1, whereas for the down-regulated, <-1 and < -2. Once it was done, the IDs of the significant genes from the GO enrichment were obtained to observe with which GO term the top three up-regulated genes were associated. It is important to note that when conducting the GO enrichment for all of the differentially expressed, the up-regulated and down-regulated genes, only the >+2 and the <-2 were used. This more restrictive filtering was done, because when using the >+1 and the <-1 the biological functions obtained with the GO enrichment tool did not seem to be related to the studied pathology.

To find correlations between the TGF- β1 and the top three differentially expressed and up-regulated genes, a list was created containing all the annotated genes associated with each GO term. In parallel, significant genes were identified using the genSel function. Each GO:ID was then labelled using a custom name format: "GOterm_" followed by a number for each gene, along with the abbreviation to indicate whether the gene was up-regulated (up) or down-regulated (down). With this information, it was possible to group all the annotated genes under their corresponding GO term and then intersect them, meaning that only the most significant genes from all of the annotated, from each GO term were obtained.

2.3.4. STRING: functional protein association networks

After the functional enrichment of differentially expressed genes was observed, the relationship between the biological processes was studied thanks to STRING.
STRING is an online dataset that allows studying the different interactions and associations between proteins. In this case, the TGF-β1 protein and the differentially expressed genes were introduced to see if there is any interaction between them.

3. RESULTS

3.1. RNA-seq analysis with Galaxy

To ensure the reliability of the data obtained through sequencing, a preliminary assessment of the quality of the reads was conducted using the FastQC tool. This analysis allowed the examination of some key quality parameters for the eight pair-end samples, as shown in **Fig. 1**. The main results obtained for each evaluated metric are detailed below, in **Annex B** further details of this analysis can be found.

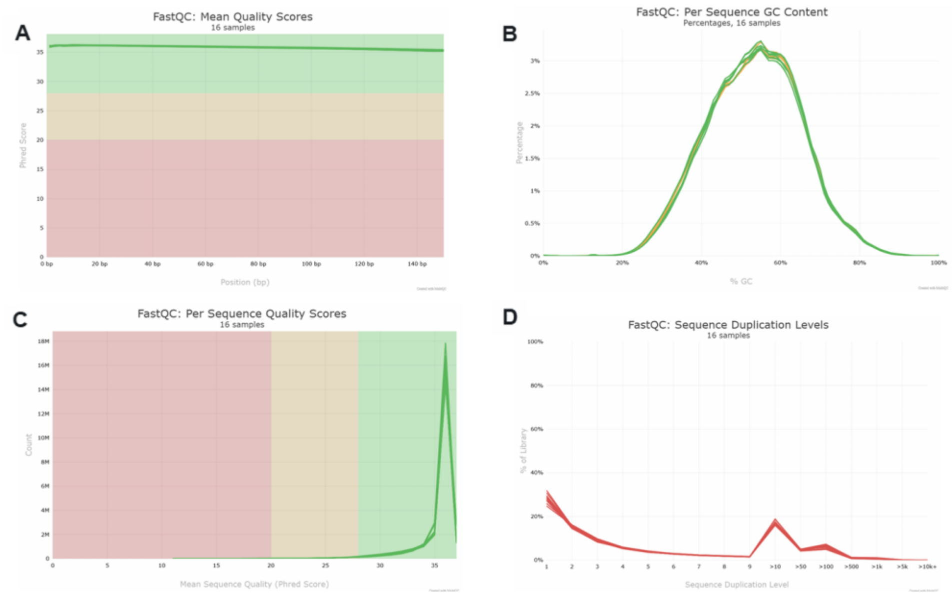


***Figure 1: Evaluation of the quality of sequencing reads for sixteen samples using FastQC. (A)** Average quality per position (Phred score) across reads, showing high and stable quality (>30) at all positions. **(B)** Distribution of GC content per sequence, with a normal curve centred around 50–55%, indicating the absence of contamination or biases in nucleotide composition. **(C)** Average quality per read, with most reads presenting values higher than thirty-five, reflecting high overall quality. **(D)** Levels of sequence duplication, with a relevant proportion of duplicate reads, due to high expression of specific transcripts or low library complexity. **(A) & (C)** Red, yellow and green areas represent low, medium, and high quality, respectively. Created with Galaxy European Server.*

3.2. RNA-seq analysis with R

3.2.1. Pre-processing and global understanding of the data

A PCA and a heatmap were generated to complete a general exploration and obtain a summarised visualisation of patterns and relationships between genes, as shown in **Fig. 2.** PCA, **Fig. 2.A,** enables clustering of samples based on their most significant variables. The two dimensions shown represent the principal components, with the accompanying percentages indicating the proportion of variance explained. The x-axis has a 44.1% of variance and the y-axis a 21.5% of variance. The shape of the samples represents the condition (Treatment or Control), and their colour indicates the individual. As a result, no clear clustering can be observed; however, a clear pattern is seen: the Treatments always appear above Controls throughout the graph.

Fig. 2.B is a representation of gene expression from normalised RNA sequencing data, with a total count matrix of 227,160 entries. Furthermore, the expression level is shown via a scale of colours (from red to blue), and the plot is divided into columns and rows: the former represents the sample, and the latter, the genes. The clustering of genes and samples is expressed by the dendrograms located above and on the left part of the heatmap. Finally, the sample annotations above the map indicate metadata variables. It is noticeable that numerous genes are not expressed at all, in the middle range of the heatmap, coloured in dark blue; however, there is a cluster in the top, coloured in yellow, that is common throughout all the samples. It is also worth mentioning that even if there is no clear difference observed between the Treatment and Control samples, there is a distinction between different Sample_IDs.

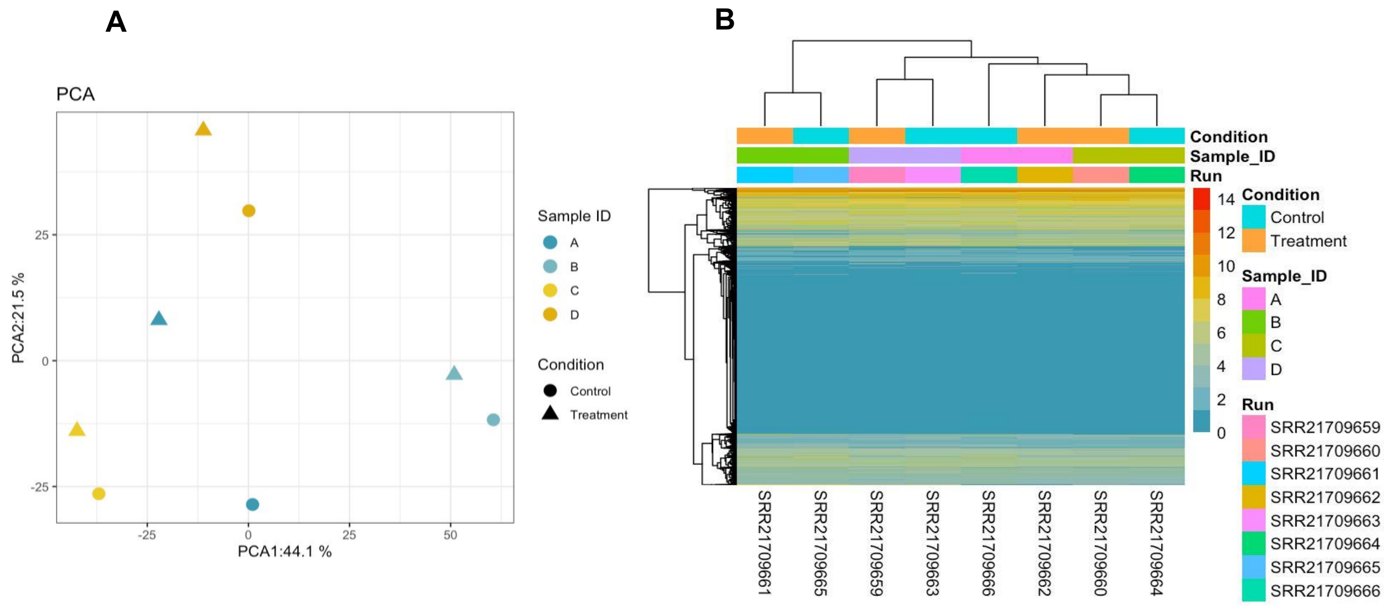


Figure 2: Principal component analysis (PCA) and heatmap of normalised transcriptome data. (A) *Samples are represented according to experimental Condition (by shape) and corresponding Sample_ID (by colour). The x and y axes show the first two components, which have 44.1% and 21.5% of the total variance, respectively. The clustering of the samples indicates a clear separation between conditions, with a quantity of variability between patients.* ***(B)*** *This plot shows a gene expression profile from RNA sequencing data normalised by applying Counts Per Million (CPM). Each column represents a sample (Run ID), and each row corresponds to a gene. The colour scale indicates the expression level, with red meaning high expression and blue representing low expression. The annotations above the samples provide information from the metadata, including Condition (Control and Treatment), Sample_ID (A, B, C and D), and Run (code of the samples). Created with RStudio®.*

3.2.2. Differential expression analysis with DESeq2

Next, a volcano plot was used to observe the most expressed genes and the differentially expressed genes in the Control vs. Treatment groups, as represented in **Fig. 3.A.** Values found under the horizontal dotted line, shown in light blue, are not statistically significant. Lighter yellow p-values are statistically significant, but their expression is low, and the dark blue p-values are from genes that are neither highly expressed nor statistically significant. The most significant gene regarding DE would be 152,816, followed by 57,611, and 3,694, all shown in dark yellow. Thanks to g:Profiler, it was possible to get the names of these three genes: ODAPH (odontogenesis-associated phosphoprotein), ISLR2 (immunoglobulin superfamily containing leucine-rich repeat 2), and ITGB6 (integrin subunit beta 6), respectively.

Fig. 3.B represents a heatmap of the statistically differentially expressed genes. As mentioned before, the expression level is shown via a scale of colours, and the plot is divided into columns and rows representing samples and genes, respectively. The sample annotations above the map indicate metadata variables. The clustering of genes and samples is expressed by the dendrograms located above and to the left part of the heatmap. In this case, a clustering of the different conditions can be observed.

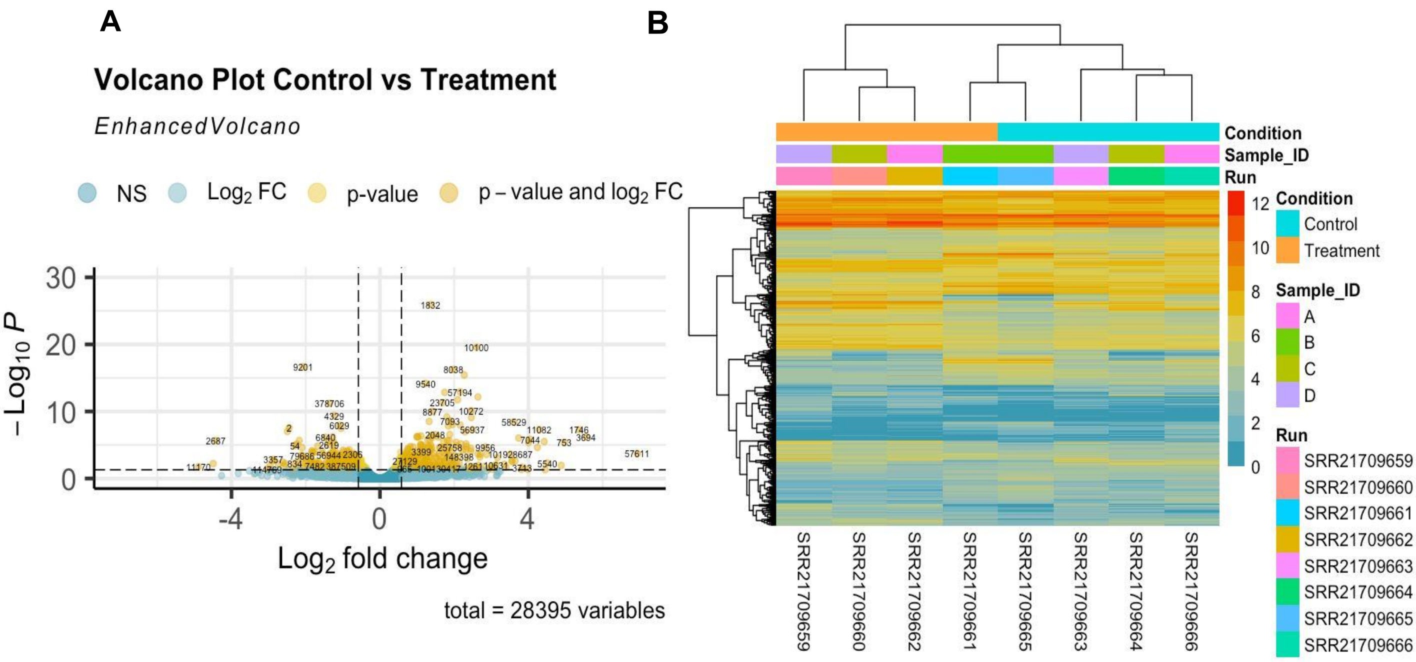


Figure 3: Volcano Plot representing Control group vs Treatment group genes and heatmap of differentially expressed genes. (A) *The x-axis represents the Log2FoldChange, and it represents the magnitude of gene expression differences; the values further to the right side are more expressed in Treatment Condition and than those on the left side of the graph are more expressed in the Control Condition. The y-axis represents the statistical significance of differential expression, the higher the value, the more significant the differential gene expression.* ***(B)*** *This plot shows a gene expression profile from differentially expressed genes, normalised by applying Counts Per Million (CPM). Each column represents a sample (Run ID), and each row corresponds to a gene. The colour scale indicates the expression level, with red meaning high expression and blue representing low expression. The annotations above the samples provide information from the metadata, including Condition (Control and Treatment), Sample_ID (A, B, C and D), and Run (ID of the samples). Created with RStudio®.*

3.2.3. GO enrichment

Regarding GO enrichment, three different enrichments were done, seen in **Table 1**, **Table 2,** and **Table 3**. In each table, the column named “Annotated” shows the number of genes from the input list associated with the GO term. The “Significant" column presents how many of the annotated genes the program has found to be statistically significant. The “Expected” column gives the number of genes expected to be included in that term by chance. Finally, the last column indicates whether the genes found in each category belong there, a low p-value means that the genes do belong in the term. It is noteworthy that there are more significant genes in each term than expected by chance.

In order to create **Table 1,** all of the differentially expressed genes were used, and it indicates that all processes are related to organism development. When it comes to **Table 2,** only the up-regulated genes were used and it can be observed that all of the functions revolve around cellular processes and their regulation. However, on **Table 3**, the functions of the down-regulated genes were analysed, and it is relevant to mention that all of the processes are involved in the regulation of other biological processes.

Table 1: GO.ID enrichment results table for DE genes. *The first column (GO.ID) is the code of the term in the GenOntology database. The term is the main function of the genes found within. The annotated column indicates the total number of genes from the input list associated with the GO term. The fourth column presents how a great deal of the annotated genes the program has found to be statistically significant. The Expected column indicates the number of genes expected to be included in that term by chance. The last column presents a p-value, if it is low, it indicates that the genes do belong to the category.*

	GO.ID	Term	Annotated	Significant	Expected	classicFisher
1	GO:0032501	Multicellular organismal process	4829	390	272.07	2.1e-19
2	GO:0048856	Anatomical structure development	4306	357	245.28	5.9e-19
3	GO:0048731	System development	3016	272	171.8	5.2e18
4	GO:0007275	Multicellular organism development	3508	300	199.82	7.9e-17
5	GO:0009888	Tissue development	1425	156	81.17	1.0e-16

Table 2: GO.ID enrichment for up-regulated genes in the Treatment samples results table by log2FoldChange>2. *The first column (GO.ID) is the code of the term in the GenOntology database. The term is the main function of the genes found within. The annotated column indicates the total number of genes from the input list associated with the GO term. The fourth column presents how a great deal of of the annotated genes the program has found to be statistically significant. The Expected column indicates the number of genes expected to be included in that term by chance. The last column presents a p-value, if it is low, it indicates that the genes do belong to the category.*

	GO.ID	Term	Annotated	Significant	Expected	classicFisher
1	GO:0007178	Cell surface protein Ser-Thr	7	7	4.03	0.0018
2	GO:0141091	TGF-β receptor	7	7	4.03	0.018
3	GO:0090092	Regulation of transmembrane receptor protein	6	6	3.545	0.033
4	GO:0006022	Aminoglycan metabolic process	5	5	2.88	0.059
5	GO:0007179	TGF-β receptor	5	5	2.88	0.059

Table 3: GO.ID enrichment for down-regulated genes in the Treatment samples results table by log2FoldChange<-2. *The first column (GO.ID) is the code of the term in the GenOntology database. The term is the main function of the genes found within. The annotated column indicates the total number of genes from the input list associated with the GO term. The fourth column presents how a great deal of the annotated genes the program has found to be statistically significant. The Expected column indicates the number of genes expected to be included in that term by chance. The last column presents a p-value, if it is low, it indicates that the genes do belong to the category.*

	GO.ID	Term	Annotated	Significant	Expected	classicFisher
1	GO:0002831	Regulation of response to biotic stimulus	4	4	1.59	0.021
2	GO0043933	Protein-containing complex organisation	6	5	2.38	0.032
3	GO:0065003	Protein-containing complex assembly	6	5	2.38	0.032
4	GO:0087435	Supramolecular fibre organisation	6	5	2.38	0.032
5	GO:0007015	Actin filament organisation	3	3	1.19	0.057

Then it was observed how the genes analysed with DE correlated with GO enrichment. All the top three genes are related to the first, second and fourth terms of **Table 1**. Regarding **Table 2,** in the third and fourth terms, none of the top three genes are expressed. However, regarding the second and fifth terms, the “Transforming growth factor beta receptor”, the ITGB6 gene is expressed. Finally, concerning **Table 3,** none of the top 3 genes are expressed in any of the terms.

3.2.4. STRING: functional protein association networks

The figures obtained from the STRING website are the ones belonging to **Fig. 4**, which showed that only ITGB6, from the differentially expressed genes, interacts with the ligand of interest.

Fig. 4.A represents the protein association networks from TGF-β1, in which direct interactions with Latent Transforming Growth Factor Beta Binding Protein 1, 3 and 4 (LTBP1/3/4) proteins as well as with its receptor (TGFR-β1 1, 2 and 3) can be seen. Other proteins that interact with TGF-β1 are Endoglin (ENG), decorin (DCN) and integrin subunit alpha V (ITGAV).

Following, **Fig. 4.B** protein association networks from ISLR2 can be observed. There are interactions with Adhesion G Protein-Coupled Receptor L1, 2, 3 (ADGRL1/2/3), Ring Finger Protein 26 (RFP26), Serine/Arginine Repetitive Matrix 3 (SRRM3), Ubiquitin Like 7 (UBL7), Netrin 3 (NTN3), Golgin A6 Family Member A (GOLGA6A), EPH Receptor A3 (EPHA3) and Coiled-Coil Domain Containing 33 (CCDC33).

Thirdly, **Fig. 4.C** shows a direct interaction between Integrin Subunit Beta 6 (ITGB6) and several proteins, including ITGAV, Integrin Subunit Alpha 2 5 (ITGA5), and TGF-β1, the protein of interest in the article.

Finally, **Fig. 4.D** indicates that Odontogenesis Associated Phosphoprotein (ODAPH) closely interacts with Family with Sequence Similarity 20 Member A (FAM20A), Ameloblastin (AMBN), Family with Sequence Similarity 83 Member H (FAM83H), G Protein-Coupled Receptor 68 (GPR68), Amelotin (AMTN), WD Repeat Domain 72 (WDR72), Enamelin (ENAM), Amelogenin, X-linked (AMELX), Solute Carrier Family 24 Member 4 (SLC24A4), and Matrix Metallopeptidase 20 (MMP20).

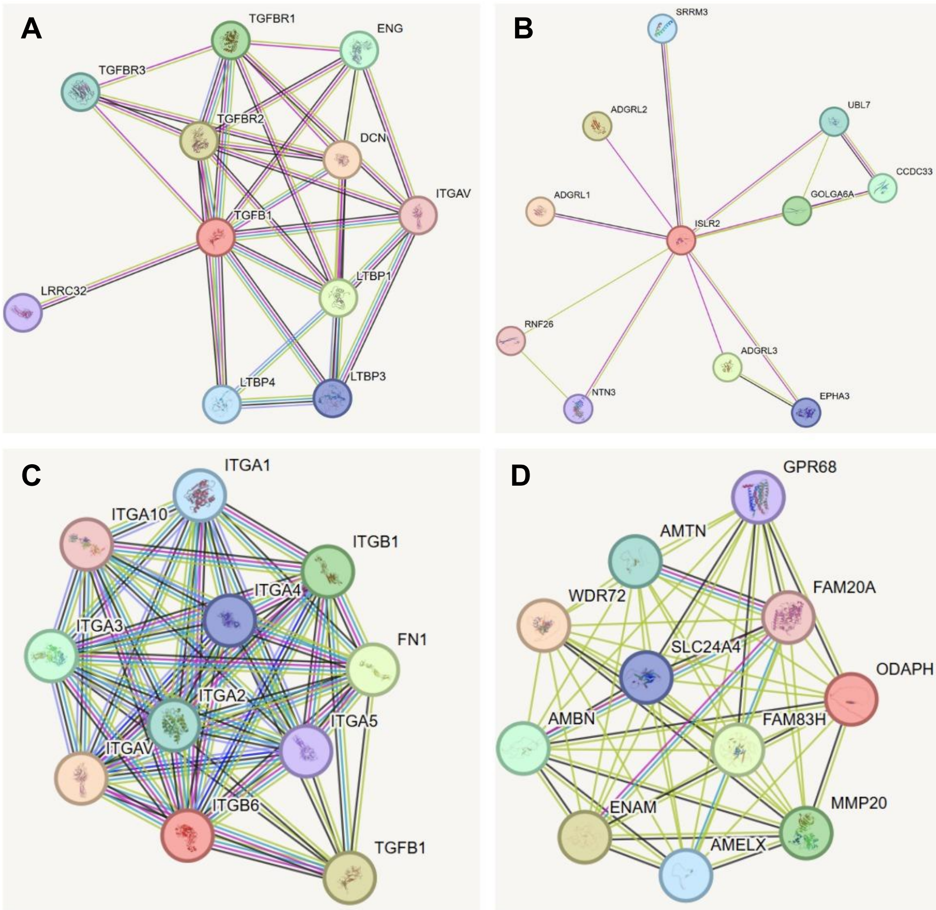


Figure. 4. TGF-β1, ISLR2, ITGB6, and ODAPH association networks generated using STRING. *Nodes represent proteins and are coloured according to their interaction stage or functional context. Edges indicate associations based on various evidence channels in STRING; these represent functional interactions and do not necessarily imply direct physical binding.* ***(A)*** *TGF-β1 network****,*** ***(B)*** *ISLR2 network,* ***(C)*** *ITGB6 network,* ***(D)*** *ODAPH network.*

4. DISCUSSION

Before discussing the results, it is important to consider the overall aim of the study: to explore how TGF-β1 influences gene expression in HTM, with the ultimate goal of elucidating the molecular mechanisms underlying XFG. Understanding these changes may help identify key genes and pathways involved in the disease, guiding future therapeutic or preventive strategies.

4.1. RNA-seq analysis with Galaxy

The graphs from FastQC found in **Fig. 1** summarise quality metrics for 16 samples. **Fig. 1.A** with the mean Quality Scores shows the average Phred quality score at each base position across the reads. All values are consistently above 30; a sequencing quality score of 30 or above is generally considered very good (CD Genomics, n.d.). There is no significant drop in quality towards the end of the reads, suggesting that the sequencing quality remains consistent throughout the read length.

Secondly, **Fig. 1.B**, in other words, Sequence GC Content, is a plot which displays the distribution of GC content across all reads. The observed curve follows the normal distribution, since it is symmetrical and consistent across samples, indicating a uniform and expected GC distribution with no signs of contamination (Khetani, 2018). To continue, **Fig. 1.C**, with the Sequence Quality Scores, shows the distribution of the average quality scores per read (CD Genomics, n.d.). Most reads have a Phred score around 35, which falls in the high-quality range.

Finally, **Fig. 1.D** shows the Sequence Duplication Levels. This plot illustrates the proportion of duplicated reads in the dataset. While a quantity of duplication is expected, especially in RNA-Seq datasets with highly expressed genes, a noticeable amount of duplication is present, particularly in low duplication levels and a few peaks at higher duplication levels. These peaks may reflect biological redundancy, a high sequencing depth or technical inaccuracies such as PCR bias or sample contamination (CD Genomics, n.d.).

4.2. RNA-seq analysis with R

4.2.1. Pre-processing and global understanding of the data

The principal objective of the plots presented in **Fig. 2** was to begin with a general exploration of the gene expression data to identify possible patterns and potential clusters between genes and samples.

PCA of gene expression data obtained by RNA-seq after normalisation and transformation using Variance-Stabilised Transformation (VST) is shown in **Fig. 2.A.** In a PCA, the first and second principal components are the x and y axes, which allow for the graphing of the data in two dimensions and identifying clusters (Frost J., 2023, January 29). In this case, the x-axis represents the Sample_ID component and the y-axis the Condition component. It can be seen that the percentage on the x-axis is higher than the percentage of the y-axis, 44.1% vs 21.5%. This indicates that the difference between the patients is greater than the difference between the Control and Treatments. That is why it is not possible to observe a clear clustering of treatments and Controls. However, samples are consistently grouped in pairs, with all Treatment samples located towards the upper region of the plot, suggesting that the data is paired. The absence of extreme outliers supports the overall consistency and quality of the dataset. This factor confirms the correct fit of the pre-processed data for subsequent differential expression analyses.

A heatmap represents the values contained in a matrix as colours (Clustered Heatmaps, 2024). In **Fig 2.B,** the most expressed genes are shown in red, and the least expressed in blue. This heatmap is a clustered one, which allows the identification of hierarchical clustering, represented as dendrograms (Clustered Heatmaps, 2024). **Fig. 2.B** suggests that there is no difference in gene expression between the Treatment and Control groups. The clustering created by the RStudio® algorithm and shown by the dendrogram reflects that there is a stronger relationship between Sample_ID rather than Condition. These results are in line with the outcome of the PCA analysis **(Fig. 2.A.)**.

4.2.2. Differential expression analysis with DESeq2

Via the volcano plot found in **Fig. 3.A,** it can be observed that gene 1,832, followed by 10,100 and 57611, are the most differentially expressed, statistically speaking. All of these 3 genes are found on the right side of the volcano plot, which indicates that these genes are more expressed in Treatment samples than in Control samples. This suggests that the genes could be related to XFG. Moreover, the 57611 gene belongs to the top three genes more expressed in the Treatment group.

To better understand their biological relevance, the top three gene identifiers were compared with their corresponding names and functions through database searches. First, gene 152,816, also known as ODAPH (odontogenesis-associated phosphoprotein), is a matrix protein involved in enamel formation, specifically during the secretory and maturation stages of amelogenesis, in other words, the process of synthesis of dental enamel or the protective outer layer of teeth (Mu et al., 2022). Next, gene 57611, which matches ISLR2, or immunoglobulin superfamily containing leucine-rich repeat 2, is involved in the positive regulation of axon extension. It is found on the cell surface and is expressed in various structures, such as the alimentary system, branchial arch, central nervous system, genitourinary system and sensory organ (Islr2 Immunoglobulin Superfamily Containing Leucine-rich Repeat 2 [Mus Musculus (House Mouse)] - Gene - NCBI, n.d.).

Finally, gene 3694 or integrin beta subunit 6 (ITGB6) encodes for a protein that is a member of the integrin superfamily. Members of this family are adhesion receptors that function in signalling from the extracellular matrix to the cell. Integrins are heterodimeric integral membrane proteins composed of an alpha chain and a beta chain. The encoded protein forms a dimer with an alpha v chain, and this heterodimer can bind ligands such as fibronectin and TGF-β1. Alternative splicing gives rise to multiple transcript variants (ITGB6 Integrin Subunit Beta 6 [Homo Sapiens (Human)] - Gene - NCBI, n.d.).

Continuing with the DE analysis, **Fig. 3.B** presents a heatmap expressing the most differentially expressed genes. A cluster of highly expressed genes appears at the top, common between the Control and Treatment samples. Dendrograms allow the visualisation of a new clustering pattern by Condition, meaning that the RStudio® algorithm does detects a clear significance in their expression.

In **Fig. 2.B**, the heatmap is organized by the Sample_ID, whereas in **Fig. 3.B,** the heatmap is organised by experimental groups, which is appropriate since the second heatmap specifically maps significantly differentially expressed genes between conditions.

4.2.3. GO enrichment

The top functions identified through GO enrichment analysis are closely associated with the development of biological processes (indicated by classicFisher value in **Table 1**). This correlation is logical, given that TGF-β1 serves as a ligand within a developmental signalling pathway (Aykul & Martinez-Hackert, 2016). On the one hand, when looking only at the enrichment for up-regulated genes in Treatment samples, it can be observed that only ITGB6 is related to the "Transforming growth factor beta receptor" term. It seems logical, since, as mentioned before, this gene is an adhesion receptor that functions in signalling from the ECM to the cell, and the XFG is related to ECM remodelling (Puthawala et al., 2007).

On the other hand, none of the top three genes are related to any of the terms from the enrichment for down-regulated genes in Treatment samples. Which makes sense, since these top three genes are the most expressed in the sample Treatment.

4.2.4. STRING: functional protein association networks

According to **Fig. 4,** TGF-β1 is highly interconnected with essential components for the control of cell growth and differentiation. On the one hand, it is important to note the direct association between ITGB6 and TGF-β1, as seen in **Fig.4.C.** ITGB6 is known to activate latent TGF-β1, which is in line with the results for the GO enrichment seen in **Table 2**, enhancing its fibrotic response (Margadant & Sonnenberg, 2010). This suggests that ITGB6 could serve as a potential biomarker or therapeutic target for managing the pathological ECM changes seen in XFG. On the other hand, **Fig.4.B** and **Fig.4.D** illustrate the networks of ISLR2 and ODAPH respectively. They do not present any interaction with TGF-β1, ITGB6 or even eachother.

Nevertheless, it is worth to note that several of the previously mentioned proteins have functions related to the disease discussed in the article. **Fig. 4.A** shows multiple proteins involved in the binding and regulation of TGFB1 activation, such as LTBP1 and LTBP3, as well as cofactors of TGF-β1 receptors implicated in angiogenesis, like ENG, or integrins involved in cellular signalling, such as ITGAV. In **Fig. 4.B**, ADGRL1, ADGRL2, and ADGRL3 are G protein-coupled receptors involved in cell adhesion. **Fig. 4.C** highlights several integrins that play a role in cell adhesion and the activation of TGF-β1. Finally, in **Fig. 4.D**, most proteins are related to dental enamel formation, although MMP20 specifically functions to degrade matrix proteins during amelogenesis.

5. CONCLUSION

Overall, it can be confirmed that the transcriptome data used for the study was reliable, as it can be observed in **Fig.1**. Having high quality data provides various benefits, such as, strengthening the credibility and validity of the work conducted, helping other researchers to replicate the study and obtain similar results and having lower risk of bias or errors.

Thanks to the DE analysis and the GO enrichment, the aim was achieved. The genes that are over-expressed in XFG and the biological processes involved in the pathology were identified. Regarding the biological functions involved, a great variety of them were found, such as, multicellular organismal process, anatomical structure development and cell surface receptor protein Ser-Thr and TGF-β receptor (**Table 1** and **Table 2**). When it comes to the over-expressed genes, only one, ITGB6 has been observed to be related to XFG since it interacts with TGF-β1. Moreover, when looking into ITGB6's function it indicates that it plays a role in ECM remodeling, which is also characteristic of XFG. Knowing this, ITGB6 could serve as a potential biomarker or therapeutic target for managing the pathological ECM changes seen in XFG.

One of the major limitations in this article is the small number of sample patients, since the study only counted with 8 different samples, which only came from 4 patients. Due to the paired nature of the data, variability was limited. Consequently, Sex and Age variables were not considered, as they did not contribute significant changes to the results. Furthermore, certain code could not be executed when accounting for these variables, as the algorithms detected correlations that interfered with analysis. Another limitation worth noting is the team's limited experience with most of the online tools and the RStudio® programming language.

For further studies, the primary recommendation is to increase the sample size as it would enhance population representativeness by incorporating greater variability, thus improving the reliability of findings. Additionally, if an animal model were to be used, as explained in **Annex A**, Mus musculus would be the best option. Furthermore, a more in-depth analysis could be conducted, where the over- and under- expressed genes are studied to determine whether a correlation or even a causal relationship exists between their expression patterns. Investigating the effects of artificially overexpressing down-regulated genes on the progression and development of XFG could provide insights for future drug development. Lastly, studying ITGB6 mechanisms and interaction may provide an alternative perspective on the XFS and XFG ivestigation.

6.DATA AVAILABILITY

The study from which the Transcriptome data was obtained can be accessed via the following bioproject ID: PRJNA882267

The Rstudio® code that was used to carry out this study is openly available in https://github.com/sanumayamf/G2_CellGrowth-differentiation_IP4.

7. AUTHOR CONTRIBUTIONS

All authors contributed equally to the introduction and conclusion. In the Materials and Methods section, Marina and Maitane focused on RNA-seq analysis with Galaxy, while Olalla and Sanu concentrated on the RNA-seq analysis with R. Each member was responsible for explaining the results of a different plot. The discussion was conducted collaboratively, focusing on how best to extrapolate the results to the genetic level; however, Olalla particularly focused on gene ontology, whereas Maitane and Sanu in DE seq analysis and Marina on the Galaxy analysis. Data analysis was led by Olalla and Sanu, with occasional support from the rest of the team. The references and final revision were supervised mainly by Marina.

8.REFERENCES

- Alexa, A., & Rahnenfuhrer, J. (2024). *topGO: Enrichment Analysis for Gene Ontology* (R package version 2.58.0). https://bioconductor.org/packages/topGO
- Aykul, S., & Martinez-Hackert, E. (2016). Transforming growth factor-B family ligands can function as antagonists by competing for Type II receptor binding. *Journal of Biological Chemistry, 291*(20), 10792–10804. https://doi.org/10.1074/jbc.m115.713487
- Blighe, K., Rana, S., & Lewis, M. (2025). *EnhancedVolcano: Publication-ready volcano plots with enhanced colouring and labeling* (R package version 1.27.0). https://bioconductor.org/packages/EnhancedVolcano
- Chen, Y., Chen, L., Lun, A. T. L., Baldoni, P. L., & Smyth, G. K. (2025). edgeR v4: powerful differential analysis of sequencing data with expanded functionality and improved support for small counts and larger datasets. *Nucleic Acids Research, 53*(2). https://doi.org/10.1093/nar/gkaf018
- Cruz-Ortega, J. S., & Boucard, A. A. (2019). Actin cytoskeleton remodeling defines a distinct cellular function for adhesion G protein-coupled receptors ADGRL/Latrophilins 1, 2 and 3. Biology Open. https://doi.org/10.1242/bio.039826
- Clustered Heatmaps. (2024). Mdanderson.org. https://bioinformatics.mdanderson.org/public-software/ngchm/heatmaps/
- Frost, J. (2023, January 29). Principal Component Analysis Guide & Example. *Statistics by Jim.* https://statisticsbyjim.com/basics/principal-component-analysis/
- Homo sapiens chromosome 19, GRCh38.p14 Primary Assembly - Nucleotide - NCBI. (n.d.). https://www.ncbi.nlm.nih.gov/nuccore/NC_000019.10?strand=2&report=genbank&from=41301587&to=41353922
- Homo sapiens isolate CHM13 chromosome 19, alternate assembly T2T-CHM13 - Nucleotide - NCBI. (n.d.) https://www.ncbi.nlm.nih.gov/nuccore/NC_060943.1?strand=2&report=genbank&from=41301587&to=41353922
- Homo sapiens isolate CHM13 chromosome 9, alternate assembly T2T-CHM13v - Nucleotide - NCBI. (n.d.). https://www.ncbi.nlm.nih.gov/nuccore/NC_060933.1?strand=1&report=genbank&from=99103647&to=99154192
- Islr2 immunoglobulin superfamily containing leucine-rich repeat 2 [Mus musculus (house mouse)] - Gene - NCBI. (n.d.). https://www.ncbi.nlm.nih.gov/gene/320563
- ITGB6 integrin subunit beta 6 [Homo sapiens (human)] - Gene - NCBI. (n.d.). https://www.ncbi.nlm.nih.gov/gene/3694
- Kanchanawong, P., & Calderwood, D. A. (2022). Organization, dynamics and mechanoregulation of integrin-mediated cell–ECM adhesions. Nature Reviews Molecular Cell Biology, 24(2), 142-161. https://doi.org/10.1038/s41580-022-00531-5
- Khetani, M. P. R. (2018, September 5). Quality control: Assessing FASTQC results. *Introduction to RNA-Seq using high-performance computing - ARCHIVED.* https://hbctraining.github.io/Intro-to-rnaseq-hpc-salmon/lessons/qc_fastqc_assessment.html

- Love, M. I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology, 15*(12). https://doi.org/10.1186/s13059-014-0550-8

- Margadant, C., & Sonnenberg, A. (2010). Integrin–TGF-β crosstalk in fibrosis, cancer and wound healing. *EMBO Reports*, *11*(2), 97-105. https://doi.org/10.1038/embor.2009.276

- MedlinePlus Genetics. (n.d.). *TGFBR1 gene.* U.S. National Library of Medicine. https://medlineplus.gov/genetics/gene/tgfbr1/
- Mu, H., Dong, Z., Wang, Y., Chu, Q., Gao, Y., Wang, A., Wang, Y., Liu, X., & Gao, Y. (2022). Odontogenesis-Associated phosphoprotein (ODAPH) overexpression in ameloblasts disrupts enamel formation via inducing abnormal mineralization of enamel in secretory stage. *Calcified Tissue International, 111*(6), 611–621. https://doi.org/10.1007/s00223-022-01023-6
- National Centre for Biotechnology Information. (n.d.). NC_000009.12:99103647–99154192 [GenBank record]. *NCBI.* Retrieved June 10, 2025, from https://www.ncbi.nlm.nih.gov/nuccore/NC_000009.12?strand=1&report=genbank&from=99103647&to=99154192
- Park, D. Y., Kim, M., & Cha, S. C. (2021). Cytokine and growth factor analysis in exfoliation syndrome and glaucoma. *Investigative Ophthalmology & Visual Science, 62*(15), 6. https://doi.org/10.1167/iovs.62.15.6
- pheatmap: Pretty Heatmaps. (2010). [Conjunt de dades]. https://doi.org/10.32614/cran.package.pheatmap
- Puthawala, K., Hadjiangelis, N., Jacoby, S. C., Bayongan, E., Zhao, Z., Yang, Z., Devitt, M. L., Horan, G. S., Weinreb, P. H., Lukashev, M. E., Violette, S. M., Grant, K. S., Colarossi, C., Formenti, S. C., & Munger, J. S. (2007). Inhibition of Integrin αvβ6, an Activator of Latent Transforming Growth Factor-β, Prevents Radiation-induced Lung Fibrosis. *American Journal of Respiratory and Critical Care Medicine, 177*(1), 82–90. https://doi.org/10.1164/rccm.200706-806oc
- Quality control: How do you read your FASTQC results? - CD Genomics. (n.d.). https://bioinfo.cd-genomics.com/quality-control-how-do-you-read-your-fastqc-results.html
- Ram, K., & Wickham, H. (2014). *wesanderson: A Wes Anderson Palette Generator* [Conjunt de dades]. https://doi.org/10.32614/cran.package.wesanderson
- TGFBR1 gene: MedlinePlus Genetics. (n.d.). https://medlineplus.gov/genetics/gene/tgfbr1/
- Tuteja, S., Zeppieri, M., & Chawla, H. (2023, May 31). Pseudoexfoliation Syndrome and Glaucoma. *StatPearls - NCBI Bookshelf.* https://www.ncbi.nlm.nih.gov/books/NBK574522/
- Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. https://ggplot2.tidyverse.org
- Aboobakar, I. F., Johnson, W. M., Stamer, W. D., Hauser, M. A., & Allingham, R. R. (2017). Major review: Exfoliation syndrome; advances in disease genetics, molecular biology, and epidemiology. *Experimental Eye Research, 154*, 88–103. https://doi.org/10.1016/j.exer.2016.11.011

9.ANNEX

9.1. Annex A: Gene sequence and protein characterisation

To obtain a first characterisation of the proteins and genes analysed in this article, various databases were used for different ends. Firstly, the genetic sequence of both receptor and ligand was analysed; they were retrieved from the NCBI nucleotide database. The receptor was referenced as NC_000009.12 *(NCBI, n.d.)* or NC_060933.1 *(Homo Sapiens Isolate CHM13 Chromosome 9, Alternate Assembly T2T-CHM13v - Nucleotide - NCBI, n.d.)*, and the ligand as NC_000019.10 (*Homo Sapiens Chromosome 19, GRCh38.p14 Primary Assembly - Nucleotide - NCBI*, n.d.) or NM_000660.7 (*Homo Sapiens Isolate CHM13 Chromosome 19, Alternate Assembly T2T-CHM13 - Nucleotide - NCBI*, n.d.).

In addition, the chromosomal location and corresponding genomic coordinates of the sequence were inspected, for which the GeneCards database was used. On one hand, the TGFβR1 gene can be found on chromosome 9, with coordinates chr9:99,103,647-99,154,192. The TGFβ1 gene, on the other hand, is located on chromosome 19, with coordinates chr19:41,301,587–41,353,922.

As a means to identify orthologous genes, BLAST (Basic Local Alignment Search Tool) was used via the NCBI BLAST website. By conducting the BLAST, it was elucidated that both the TGF-β1 and its receptor come from *Homo Sapiens*. It is worth noting that it can also be found in other species, such as *Mus musculus* (mouse), with which the receptor had an 84.02% similarity and the Ligand an 81.74% similarity.

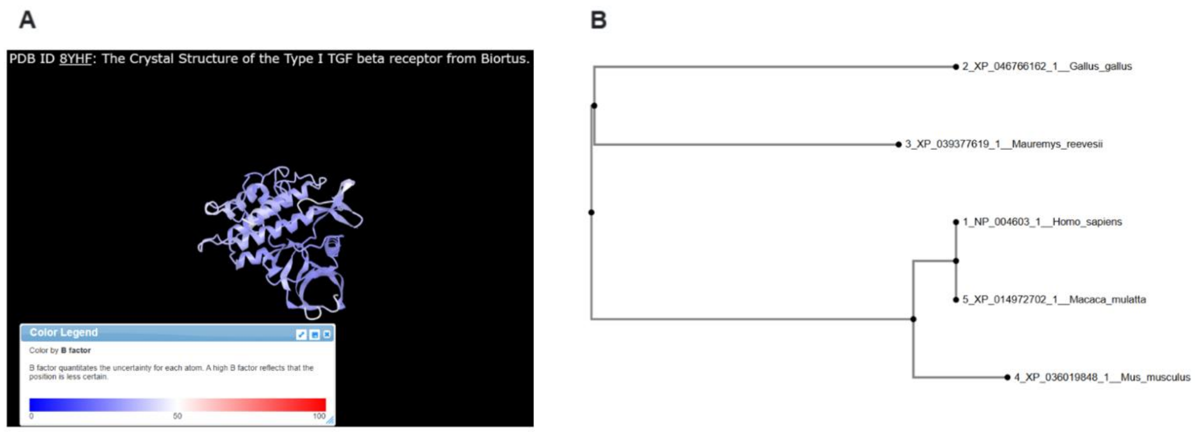
Following this initial step, the focus shifted from nucleotide sequences to protein-level analysis. Protein domains and motifs were examined using the NCBI, UniProt, and Ensembl databases. Regarding TGFβ1R, it contains 503 amino acids, and its ligand contains 391 amino acids. Concerning protein domains, the receptor has 2 domains, the GS domain and the Protein kinase domain. It also contains an FKBP1A-binding motif. In the case of the ligand, it only has a single domain, the cell attachment site.

Next, the structural analysis was examined, and the 3D structures of the proteins were retrieved and evaluated based on resolution and B-factor values. The Cn3D visualisation tool was then used to analyse and colour the structures based on the B-factor data, as shown in **Supl.Fig 1.A** The protein chosen for this process was the one with the lowest resolution, 1.40 Å (8YHF); the PDB database was used. In this case, most of the protein appears blue, indicating low B-factors, which suggests that the protein structure is stable and reliable.

Multiple Sequence Alignment (MSA) was performed using MAFFT to obtain an output for studying the evolutionary relationships between the different sequences. To obtain the MSA, a multifasta file with the orthologous sequences at the protein level was introduced to MAFFT. The species chosen were the *Macaca mulatta, Mus musculus* and other random species that could represent different animal classes. The first two were chosen so that an experimental animal could be chosen in case more analysis was needed. To create the phylogenetic tree, Phylo.io was used, a tool included in MAFFT.

According to Ensembl, the TGFβR-1 gene has 20 transcripts (splice variants). These transcripts differ in their length and exon composition. Not all of them code for proteins; a few are nonsense-mediated decay, and others retain intron transcripts that may play regulatory roles. The database reports 227 orthologues, one of them being the *Mus musculus;* at the protein level, the alignment shows a 96.42% identity, indicating a high level of conservation between the two species. Since both; the receptor and the ligand; originate from *Homo sapiens*, it is hypothesized that primates and other closely related mammals will be more similar in the phylogenetic tree. In contrast, species such as turtles (*Mauremys reevesii*) and birds (*Gallus gallus*) are expected to be more distantly related, as shown in **Supl.Fig 1.B.**

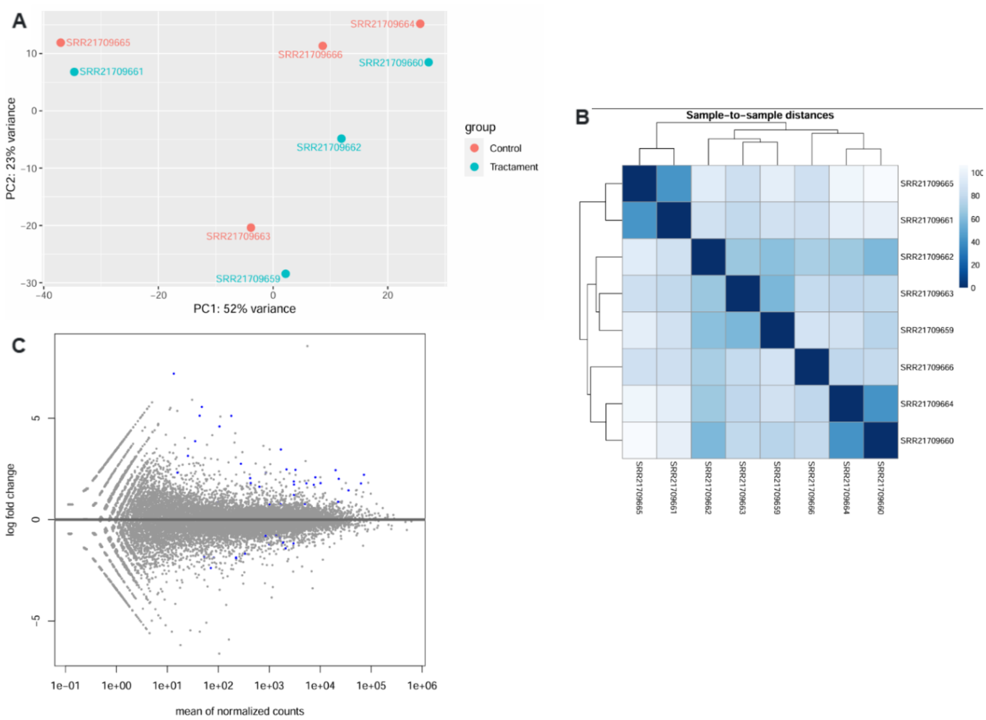
Regarding the orthologues, the most suitable model for further studying the TGFβR-1 pathway in future studies would be Macaca mulatta, as it is the closest species to Homo sapiens in the phylogenetic tree. However, due to ethical and economic constraints, Mus musculus is a more practical choice. Despite being more distantly related, mice share significant genetic homology with humans. Moreover, Mus musculus is easier to maintain under laboratory conditions, is more cost-effective, and is one of the most extensively characterised models in biomedical research.



Supplementary Figure 1: Protein characterisation. (A) *The crystal structure of the Biortus TGF beta type I receptor, stained for B factor, shows predominantly blue, indicating low B factors, suggesting that the protein structure is stable and reliable.* ***(B)*** *The phylogenetic tree shows the distance between species, showing the poricity between Macaca mulatta, and Homo sapiens. Gallus gallus and Mauremys reevesii could be considered ancestral*

9.2. Annex B: Quality control assessment

Once the quality of the sequencing data was confirmed, the gene expression analysis performed is presented in **Supl.Fig. 2**.



Supplementary Figure 2: Analysis of gene expression data using DESeq2 in Galaxy. (A) *Principal component analysis (PCA) shows the clear separation between "Control" (red) and "Treatment" (blue) groups, indicating a significant variation in the transcriptomic profile.* ***(B)*** *Distance matrix showing the similarity between samples based on Euclidean distance, showing coherent clustering according to treatment.* ***(C)*** *MA plot (log2 fold change vs mean normalised counts), where the blue dots indicate genes that are significantly differentially expressed between the two groups. Created with Galaxy European Server*

Concerning **Supl.Fig.2**, specifically, **Supl.Fig.A**, the variance explained by the first two components (PC1: 52%, PC2: 23%) adds up to 75%, indicating that most of the variability between samples is represented here. It can be seen that the percentage on the x-axis is higher than the percentage of the y-axis, 52% vs 23%. This indicates that the difference between the patients is greater than the difference between the Control and the Treatments. That is why it is not possible to observe a clear clustering of Treatments and Controls. However, samples are consistently grouped in pairs, with all treatment samples located toward the upper region of the plot, which suggests that the study works with paired data

Supl.Fig.2.B shows the sample-to-sample distances, revealing a symmetrical pattern along the diagonal, as the distance between two samples A and B is equal to that between B and A.

The diagonal, which represents the distance of each sample from itself, is zero and appears in the darkest colour. Hierarchical clustering highlights coherent clusters within individuals, rather than experimental conditions, for example, SRR21709660 and SRR21709664 (same patient) have a smaller distance than SRR21709666 and SRR21709664 (both Control, different patients), which supports the PCA outcome.

Finally, **Supl.Fig. 2.C** or MA plot shows how the genes are distributed according to the change in expression (log fold change) for their average abundance (mean of normalised counts).
The blue dots represent genes with statistically significant differential expression, a number of which show a log2 fold change (log2FC) greater than 1 or less than -1, indicating the difference between Controls and Treatment samples.

This threshold is used instead of 0 to highlight genes with biologically meaningful changes, specifically, those whose expression has at least doubled or been reduced by half. As expected, most genes are clustered around a log2FC of 0, showing no major change, but a notable portion displays significant shifts in expression.